Voltera: PV Detection from Electricity Consumption

Finding solar installations in the patterns of everyday demand

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1 Time-series problem and data

We built Voltera to answer three questions from a customer's electricity history: is there a photovoltaic (PV) installation, when did it first come into service, and what is its nominal capacity? The AIL challenge gave us 15-minute grid-withdrawal measurements, 40 labelled development customers and 1,148 additional unlabelled histories. Another 20 customers were reserved for hidden testing [1]. We could use the timestamps, but not weather, irradiation or location data.

Our starting point was the effect of solar generation on the meter. Once panels supply part of a building's daytime demand, it needs less electricity from the grid. Depending on the meter, surplus production can appear as negative readings or simply as zero imports. Neither gives us a separate measurement of generation. We therefore looked for a change in the shape of demand, rather than treating a low midday reading as proof of PV.

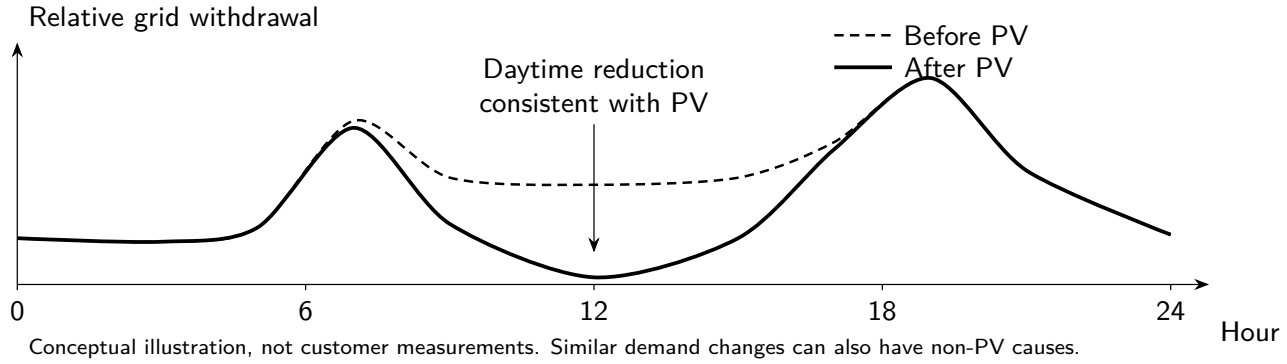


Figure 1: We looked for a persistent daytime reduction while using the rest of the day as a reference. These curves illustrate the idea, not a measured customer or a prediction.

That distinction matters because ordinary consumption already has considerable structure. Morning and evening routines create daily peaks, working days differ from weekends, and demand changes with the seasons. A springtime reduction can resemble a new solar installation. Conversely, a system installed in winter may not leave a strong daytime signature until later. Holidays, occupancy changes and batteries make the interpretation harder still. We wanted to find a lasting, solar-like change against this background, rather than forecast the next reading.

Before modelling, the loader validates the input schema, orders readings by epoch timestamp and drops duplicates. We kept missing readings separate from measured zeros, since filling a midday gap with zeros would create exactly the signal we were trying to detect. Epoch time also resolves daylight-saving transitions: the submitted pipeline places every reading on a fixed UTC+1 grid of 96 slots per day, so the 92- and 100-interval local days need no special case and PV keeps solar time. Coverage checks keep sparse days and weeks from dominating the evidence.

We also kept the roles of the datasets separate. An absent label did not make a customer PV-negative, so the unlabelled histories entered only our Transformer research and a pseudo-labelling comparison that inner selection rejected; the submitted model is trained on the 40 labelled customers alone. Synthetic customers let us test installations, gaps and demand changes under known conditions. We distinguish these diagnostics from aggregate development evaluations and include no real-customer traces.

2 Time-series methodology

We combined two views of the same history: how daytime demand changes relative to the customer's other consumption, and whether weekly profiles resemble those of PV customers. Date and capacity estimates stay tied to explicit temporal and physical features rather than coming from one model for all three answers.

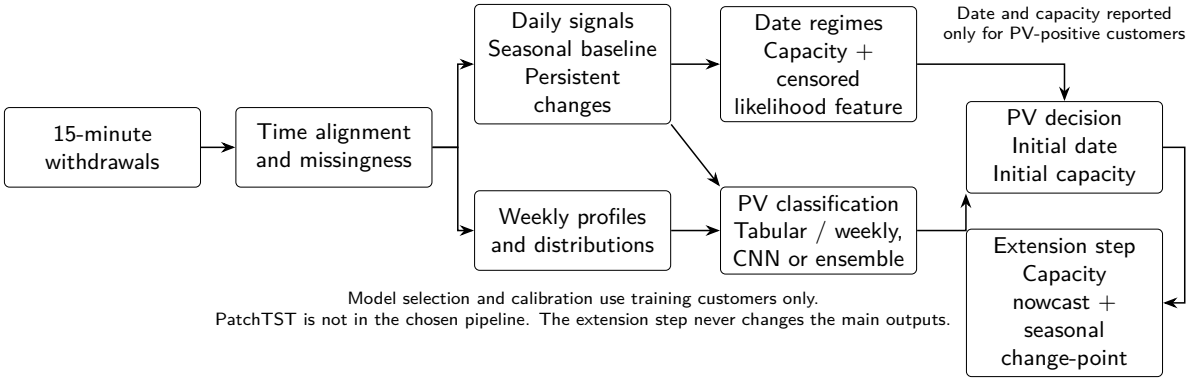


Figure 2: We use daily signals for physical evidence and weekly profiles for learned presence detection. Date and capacity estimates are reported only for customers classified as having PV; the extension step then runs on those customers without changing the main predictions.

For the physical view, we summarized midday, night and evening consumption, their relative levels, the share of zero readings and the duration of the zero-import window around noon. Night and evening demand serve as references: a reduction across the whole day means something different from a change concentrated around midday. Export features from negative readings are enabled only when enough training customers show them; they are off in the submitted model.

We then removed typical seasonal variation from the midday/night log-ratio using a week-of-year baseline fitted on non-PV training customers. Binary segmentation proposed change points on seven-day summaries, followed by local refinement. We checked whether a drop persisted and whether similar changes recurred in other years. Dating favors the earliest sufficiently supported onset, while classification keeps its standard candidate features. These checks separate a new installation from a recurring seasonal pattern, though not every demand change.

For presence detection, we implemented logistic regression, shallow CatBoost and weekly-evidence models. We also adapted the CNN architecture of Neubert et al. [2], combining a 168-hour profile with a 100-point consumption-density representation. We normalized both inputs per customer, trained the branches separately and fitted a combined head with frozen branches. Window weights give each customer equal total contribution. We summarize CNN evidence using the weakest probability in the strongest run of four consecutive covered weeks. This preserves recent evidence, but a history without four such weeks scores zero and cannot then pass the ensemble threshold. We also added an ensemble that averages the CNN probability with the selected tabular/weekly classifier. Shared inner folds select its threshold and compare it with the other candidates, so enabling the CNN does not force its adoption; the submitted fit on all 40 customers chose this ensemble over a CatBoost base at threshold 0.5.

For dates, we distinguish a visible event, PV present at the start, and insufficient evidence, with training-only calibration. For capacity, we combine midday changes, load levels and zero-window width with Ridge or CatBoost regression of log-capacity and a fitted scale factor; the submitted fit selected Ridge. We later added a Tobit-style censored-likelihood estimate: a zero-import reading tells us that generation covered demand, not that generation was zero. Pre-onset load per time slot is lognormal, adjusted for night-load changes; PV follows a timestamp-only solar shape times the training-fitted seasonal curve. Positive imports contribute a likelihood density, near-zero readings a probability bound, and the resulting log-capacity is an additional regression feature, not a production measurement. It needs enough pre-onset and April–August post-onset days, stays missing otherwise, and is always included, with an explicit opt-out. This follows the censored-observation idea of Brown et al. [3], with lognormal load and timestamp-only solar proxies rather than their full model.

For extensions, a separate step runs after the main model on the customers it classifies as PV. A CatBoost regressor fitted on the training customers' labelled capacity regimes nowcasts installed log-capacity from 35-day windows of zero-import run lengths and load levels. A per-customer change-point model fits the customer's own annual harmonics jointly with up to two persistent steps of at least 15% and $z \geq 4$; the date comes from a two-segment fit of the year-over-year zero-run envelope, the capacity after the event from the new plateau, and at most three events are reported. All thresholds are fixed constants from synthetic seeds, and the main predictions never change. We implemented the work in Python using NumPy/pandas, SciPy, scikit-learn, ruptures, CatBoost and PyTorch.

3 Experimental results and discussion

With only 40 labelled customers, we needed to be careful about what counted as independent evidence. We split by customer, keeping each customer's windows together. Inner folds select configurations and thresholds; pooled outer predictions assess the procedure. We used shared splits for comparisons, following reproducible time-series benchmarking practice [4]. We also logged design changes informed by outer results. An audit later included lock-box customers, so we cannot describe subsequent evidence as a clean independent lock-box evaluation.

The challenge score is $S = F_1(70 + 15D + 15P)$. Date reward falls to zero at 30 days of error, and capacity reward at 50% absolute percentage error. Both are averaged only over correctly detected PV customers. We therefore consider detection and conditional errors together: changing which customers are detected can change these errors without changing the estimators. The results below are development comparisons, not hidden-test scores. Repeated folds and bootstrap intervals do not create additional independent customers or capture all selection uncertainty.

Our team's decision log records adoption of the CNN probability ensemble after a paired development comparison on all 40 customers, lock-box included, gave a score increase of 4.06 points, with a 90% interval of [1.40, 7.56]. Pseudo-labelling was not selected in any of the 20 outer fits in its comparison. A subsequent pre-registered capacity comparison used three fresh customer-split seeds on all 40 development customers. Adding the fixed censored-likelihood feature raised capacity reward from 0.419 to 0.537, with a paired total-score increase of 1.44 points and interval [0.45, 2.42]; detection and dates were identical. These are recorded team evaluations, not runs repeated for this report. Fresh splits still reuse the development customers, so we do not treat them as a new independent dataset.

The extension detector was developed on synthetic data only: 15 customer cases, nine with labelled extensions and six look-alikes, scored end to end from the main model's own predictions. On a fresh seed block, the shipped harmonic detector raised the mean bonus of a Monte Carlo hidden-test geometry from 0.60 to 1.11 points (variant A) and 0.92 to 1.52 (variant B), but false alarms per draw rose from 0.66 to 0.77 and 0.61 to 0.95, missing its pre-registered margin; we shipped it by an explicit override because the bonus cannot fall below zero. The pre-registered real-data gate, nested cross-validation on the 40 development customers, gave 0.50 false extensions per outer repeat, under the 1.00 limit, but did not recover the single labelled development extension. Small, late and second extensions remain the weak spots.

We also tried to teach PatchTST the "language" of electricity time series [5], with four-hour patches as words and a week as their context. By hiding readings and reconstructing them, we could learn from unlabelled histories before applying PV labels. We trained small and medium encoders, then explored frozen features, limited fine-tuning and a CNN-gated hybrid. The difficulty was turning reconstruction into the answers we needed. Reconstructed-minus-observed demand was not established as PV generation, and learning typical consumption did not directly teach installation dates or nominal capacity. Without a reliable downstream benefit to justify the complexity, we moved to task-specific CNN detection and physical/statistical estimation. PatchTST is not in the chosen Voltera pipeline.

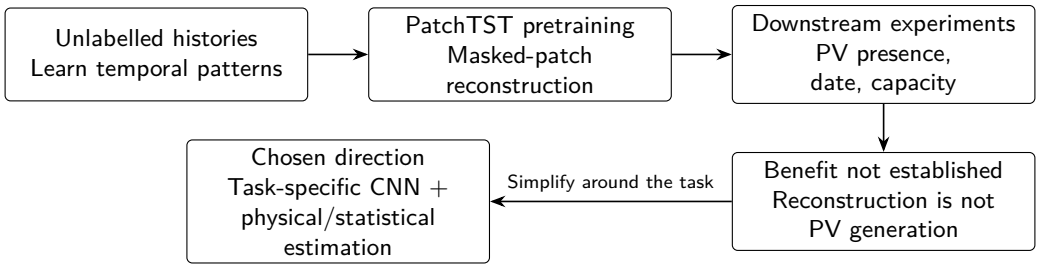


Figure 3: Our Transformer work taught us to distinguish learning consumption patterns from answering the PV tasks. We retained the trained encoders for research but did not include PatchTST in the chosen pipeline.

Alongside the models, we built offline inference, exact CSV outputs, serialization, hygiene checks and a local GUI. We used LLM coding assistants for implementation, tests, debugging and drafting, but no LLM runs inside training or inference. We remain responsible for checking the results and the content of this report.

References

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